

Limits

Let f be a function that is defined on an open interval containing a , but possibly not defined at itself. Let L be a real number. The statement $\lim_{x \rightarrow a} f(x) = L$ defines the limit of the function $f(x)$ at a point a . L is the value that the function has as the point a is approached. *Excel Review Center*

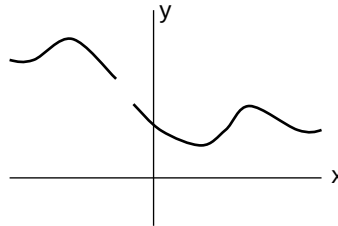
Theorems of Limits

The following are important theorems and properties of limits considering the following: $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = K$

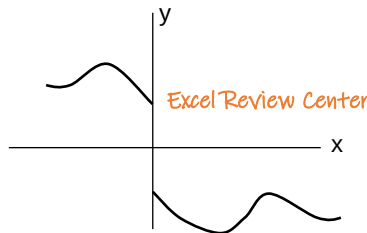
1. If $\lim_{x \rightarrow a} f(x)$ exists then it is unique.
2. $\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$
 $\lim_{x \rightarrow a} [f(x) + g(x)] = L + K$
3. $\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$
 $\lim_{x \rightarrow a} [f(x) - g(x)] = L - K$
4. $\lim_{x \rightarrow a} [f(x) \cdot g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$
 $\lim_{x \rightarrow a} [f(x) \cdot g(x)] = L \cdot K$
5. $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{L}{K}$ ($K \neq 0$)
6. $\lim_{x \rightarrow a} \frac{1}{g(x)} = \frac{1}{K}$ ($K \neq 0$) *Excel Review Center*
7. $\lim_{x \rightarrow a} [f(x)]^n = \lim_{x \rightarrow a} [f(x)]^n$ for $n > 0$
8. $\lim_{x \rightarrow a} [cf(x)] = c \left[\lim_{x \rightarrow a} f(x) \right]$
9. $\lim_{x \rightarrow a} cx^n = c \lim_{x \rightarrow a} x^n = ca^n$
10. If f is a polynomial function, then $\lim_{x \rightarrow a} f(x) = f(a)$ *Excel Review Center*
11. $\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)}$
when n is a positive integer
12. If $f(x) \leq h(x) \leq g(x)$ for all x in an open interval containing a , except possibly at a , and if $\lim_{x \rightarrow a} f(x) = L = \lim_{x \rightarrow a} g(x)$ then $\lim_{x \rightarrow a} h(x) = L$.

One-Sided Limits

Suppose f is a function such that it is not defined for all values of x . Rather, it is defined in such a way that it "jumps" from one y value to the next instead of smoothly going from one y value to the next.



The figure above shows that $y = f(x)$ is not defined for all values of x .



The figure above shows that $y = f(x)$ "jumps" from a positive value to a negative value.

The statement $\lim_{x \rightarrow a^+} f(x) = R$ means that as x approaches " a " from the right or from positive infinity, the function f has the limit R .

Likewise, the statement $\lim_{x \rightarrow a^-} f(x) = L$

means that as x approaches " a " from the left hand side or from negative infinity, the function f has the limit L .

If f is defined in an open interval containing a , except possibly at a , then $\lim_{x \rightarrow a} f(x) = L$ if and only if

$$\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x)$$

CONTINUITY

A function f is continuous at a point a if the following are the three conditions being satisfied:

1. $f(a)$ exists, that is, f is defined at a ,
 2. $\lim_{x \rightarrow a} f(x)$ exists, *Excel Review Center*
 3. the two numbers are equal.
- Continuity test at a point $x = a$:
- $$\lim_{x \rightarrow a^-} F(x) = \lim_{x \rightarrow a^+} F(x) = F(a)$$

Types of Continuities

1. **Jump Continuity** – occurs when the curve "breaks" at a particular place and starts somewhere else.
2. **Point discontinuity** – occurs when the curve has a "hole" in it from a missing point because the function has a value at that point that is "off the curve".
3. **Essential discontinuity** – occurs when the curve has a vertical asymptote. *Excel Review Center*
4. **Removable discontinuity** – occurs when there is a rational expression with common factors in the numerator and denominator.

Special Limits

1. $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$
2. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$
3. $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$
4. $\lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n = e$ *Excel Review Center*
5. For $a > 1$, then:

$\lim_{x \rightarrow +\infty} a^x = +\infty$	$\lim_{x \rightarrow -\infty} a^x = 0$
$\lim_{x \rightarrow +\infty} \log_a x = +\infty$	$\lim_{x \rightarrow 0^+} \log_a x = -\infty$
6. For $0 < a < 2$, then:

$\lim_{x \rightarrow +\infty} a^x = 0$	$\lim_{x \rightarrow -\infty} a^x = +\infty$
$\lim_{x \rightarrow +\infty} \log_a x = -\infty$	$\lim_{x \rightarrow 0^+} \log_a x = +\infty$



The term "Calculus" was derived from a Latin word "calx" which means "stone" and from a Greek word "chalis" which means "limestone". *Excel Review Center*

In 1684, a German mathematician and philosopher Gottfried Wilhelm von Leibniz published his early work on calculus, while an English astronomer, physicist and mathematician Isaac Newton made an early study on the subject in 1665 but did not published his work until 1704. These two mathematicians are now considered as the founders of today's calculus.

The subject Calculus is divided into four areas namely, differential calculus, integral calculus, differential equations and calculus of variation. Differential Calculus is a branch of mathematics which deals with derivatives and limits.



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